

MyDesign Portfolio

Element G



	Details	Tools and Manufacturing Methods used
Step 1	2 rooms is 12x12 and 6x12, which is gonna total to an 18x12 structure	Cardboard, hot glue gun, cutting tools, and rulers. You have to cut all the pieces and use the hot glue gun to connect them
Step 2	There must be one doorway that is 6.5x2.5 that leads out the front door, and a doorway the same size connecting the two rooms	Cardboard, hot glue gun, hinges, you have to use the hinges to make the door open and close, and have a doorway inside the house connecting the two rooms

Step 3	Make furniture that resembles a 2-bed, 2-desk, and 2-chair	foam plastic, cotton, wood, and cloth
Step 4	Make a detection and notification system that detects the fire in 30 seconds or less	Arduino, breadboard, LED lights, infrared sensor, wires, buzzer, water pump
Step 5	The Suppression systems have to take the fire out in less than 5 minutes	Water pump, tubings, battery
Step 6	The cost of the entire project has to be \$50 or less, including the value of donated materials	Find materials on Amazon and eBay
Step 7	The Removable roof must be able to come off, but it has to fit on well, so the interior is able to be inspected	Cardboard, glue, and cutting materials



	Details	Tools and Manufacturing Methods used
Step 1	2 rooms is 12x12 and 6x12, which is gonna total to an 18x12 structure	Metal pillars, walls, screws, and razors

Step 2	There must be one doorway that is 6.5x2.5 that leads out the front door, and a doorway the same size connecting the two rooms	Walls, cutting razors
Step 3	Make furniture that resembles a 2-bed, 2-desk, and 2-chair	Carpet, foam, sticks, paper, glue
Step 4	Make a detection and notification system that detects the fire in 30 seconds or less	Arduino, breadboard, LED lights, infrared sensor, wires, buzzer, water pump
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Material name	Description	Where the material was purchased	Cost per unit	# units	Total cost per item
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Buzzer	It will let out a sound when it detects a fire	Amazon	\$1 per unit	6	\$7
Audriono	You put code into it, and it allows the buzzer and the infrared sensor to work	Amazon	\$27.60	1	\$27.60
breadboard	Allows the wires to connect to the arduino and	Amazon	\$1.50	4	\$6.79
wires	To help connect the breadboard and the Arduino, to put everything together	Amazon	\$2.50	3	\$6.98
LED lights	It helps us know that the systems are working	Amazon	\$1.25	50	\$3.97
Infrared sensor	It senses the amount of heat and smoke inside of the home and sends a signal to the buzzers and the water pump	Amazon	\$1.75	10	\$8.79

Water pump and tubing	When the water pump is activated, we poke holes into our tubes to allow the water to shoot out and allow the water to flow and extinguish the fire	Amazon	\$2.15	4 water pumps and 1 meter tubing	\$13.99
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There will be two rooms: one 12"x12", and one 6"x12". The ceiling MUST be removable, though when put in place should fit reasonably well. The interior must be accessible for inspecting the furniture and for igniting the fire.	This ties in to my prior research because when finding information about effective fire detection systems, I found out that the structure of a building plays an important role
There must be one doorway (6.5" x 2.5") leading out of the front room of the compartment (to outside) and one doorway of the same size	With some prior research, we found that doorways allow for more oxygen to escape. The number and size of openings in a building significantly influence fire behavior, impacting ventilation, fire spread, and structural stability.

connecting the two rooms by means of the interior wall. There must be one window on two exterior walls for the larger room (two windows).	
Each team must build and furnish their room with the following furniture. The furniture must be representative of real furniture, i.e. materials selected for furniture must be similar to materials used in actual homes and incorporate materials such as foam plastic, cotton, wood, and cloth.	This ties in to our prior research about fuel sources, because materials like cotton and plastic will increase the fire magnitude.

The system must be able to detect and activate the notification systems about a	This ties in to my prior research because when finding information about detection systems, I found that notification time in fire incidents is crucial for effective response and safety. Timely alerts allow individuals to evacuate, first responders to arrive quickly, and fire departments to manage the situation

fire in less than 30 seconds.	efficiently. Fast response times significantly reduce the risk of injuries and property damage.
Time for the fire to be controlled by the suppression system. Must be done in less than 5 minutes.	This ties in to our prior research because when finding information about suppression systems, I found that time is a crucial factor in fire suppression. Early and effective suppression can prevent a fire from spreading and becoming larger, more intense, and more destructive.
The total estimated cost must be under \$50 dollars, including the value of any donated materials. By imposing a budget, it gives a plan to follow and allows for fairness between all designs	

The ceiling MUST be removable, though when put in place should fit reasonably well. The interior must be accessible for inspecting the furniture and for igniting the fire.	This ties in to our prior research because when finding information about house fires, I found that a building's roof plays a crucial role in fires by influencing their development and impact, particularly in wildfires.
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Attribute	Description	Describe your prototype's functionality to test this attribute
Metal frame and model house	A model home that uses a flammable ceiling that holds back a mixture of sand and baking soda to help put out a fire. This system will use Arduino and temperature sensors to activate a buzzer in the event of a fire.	

Detection system and notification	Using an Arduino and an MQ2 smoke detector and IR to activate a buzzer for notification.	Using Supervised fires near our detection system and changing variables like the presence of smoke and just an infrared source or just smoke and no infrared light source to ensure that there are no false alarms and to help set thresholds for our smoke detectors.
Suppression	To minimize false alarms, we will employ two sensors for fire detection: an MQ2 gas sensor and an infrared (IR) sensor. A buzzer will be utilized for notification purposes.	We will also use controlled fires for our detection system and allow our suppression system to activate with the notification system.

	Additionally, we will install tubes around the entire perimeter of the building to transport water throughout the entire building to put out a fire .	
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Attribute	Justification for why this attribute cannot be modeled or tested with your prototype
Testing the flammability of the furniture	We wanted to test how quickly the our furniture would quickly go into flames and we weren't able to test it because we didn't have enough time and only had one design
Testing the Detection and suppression system	We wanted to test how good and fast out detection and suppression systems is, but we didn't have enough time or enough resources to test them